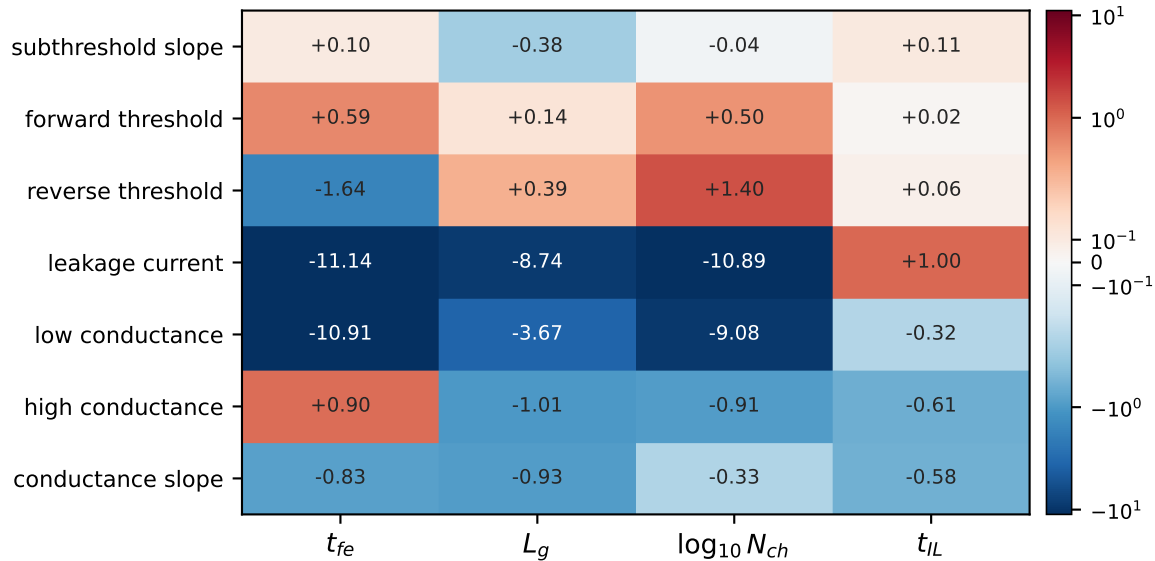


(a) the manufactured Jacobian, $d\log(\text{FoM})/d\theta$ 

(a) is 2D+1 = 9 central-difference probes of DEVSIM at the flagship's starting corner, all served from results/cache/devsim/.

Relative units, because the seven figures of merit are measured in volts, amps, siemens and mV/dec. (b) is measured along the flagship's own accepted path: anchor a true Jacobian, patch it by Broyden from the secant pair each step supplies free, and rebuild the true one alongside. The grey bars are the trust-region step length, and they are why the curve recovers after $k=6$ -- a rank-one patch over a step of 0.02 barely has to be right. Panel (a) calls no solver; panel (b) cost 11 DEVSIM probes on top of the 70 already cached.

(b) the free patch goes stale with DISTANCE, not with steps

